Total No. of pages: 2 *Roll No.:..........................*

B.TECH.– FOURTH SEMESTER (Backlog/Improvement)
MID-SEMESTER EXAMINATION, February 2025

Course Code-COMTC13/CAMTC13/CBMTC13/CDMTC13/CIMTC13
Course Title-Probability and Stochastic Processes
Time: 1:30 hours Max Marks-25

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

<table>
<tr><th>Q. No.</th><th>Question</th><th>Marks</th><th>CO</th></tr>
<tr><td>1(a)</td><td>In a college, 25% of boys and 10% of girls are studying mathematics. The girls constitute 60% of the student body. If a mathematics student is selected at random, what's the probability that the student is (a) a girl, (b) a boy?</td><td>2.5</td><td>CO1</td></tr>
<tr><td>1(b)</td><td>A random variable X has the following distribution:
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr>
<tr><td>$P(x)$</td><td>0</td><td>k</td><td>$2k$</td><td>$2k$</td><td>$3k$</td><td>k^2</td><td>$2k^2$</td><td>$7k^2 + k$</td></tr>
</table>
(i) Find k.
(ii) Evaluate $P(X < 6), P(X \geq 6)$ and $P(0 < X < 5)$.</td><td>2.5</td><td>CO1</td></tr>
<tr><td>2(a)</td><td>A probability curve $y = f(x)$ has a range from 0 to ∞. If $f(x) = e^{-x}$, find the mean and the third moment about the mean.</td><td>2.5</td><td>CO1</td></tr>
<tr><td>2(b)</td><td>Define kurtosis of a probability distribution. Suppose the first four moments of a distribution are $1, 4, 10$ and 46 respectively. Compute the coefficient of kurtosis and define its nature.</td><td>2.5</td><td>CO1</td></tr>
<tr><td>3(a)</td><td>Suppose a continuous random variable X has the probability density $f(x) = \begin{cases} k(1-x^2) & 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$
(a) Find k.
(b) Calculate the mean and variance.</td><td>2.5</td><td>CO2</td></tr>
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<table>
<tr><td>3(b)</td><td>If X is having uniform distribution with mean 1 and variance $\frac{4}{3}$. Find $P(X < 0)$.</td><td>2.5</td><td>CO2</td></tr>
<tr><td>4(a)</td><td>After correcting 50 pages of a book, the proof reader finds that there are, on the average, 2 errors per 5 pages. How many pages would one expect to find with $0, 1, 2, 3$ and 4 errors in 1000 pages of the first print of the book?</td><td>2.5</td><td>CO2</td></tr>
<tr><td>4(b)</td><td>A coin is weighted so that its probability of landing on heads is 20%. Suppose the coin is flipped 20 times. Find a bound for the probability that it lands on heads atleast 16 times.</td><td>2.5</td><td>CO2</td></tr>
<tr><td>5(a)</td><td>Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys, (b) 5 girls, (c) either 2 or 3 boys?</td><td>2.5</td><td>CO2</td></tr>
<tr><td>5(b)</td><td>Define multionomial distribution in brief.</td><td>2.5</td><td>CO2</td></tr>
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